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Towards Columbia: Paleomagnetism of 1980–1960 Ma Surumu volcanic rocks, Northern Amazonian Craton

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ABSTRACT

We present a paleomagnetic study on the well-dated 1980–1960 Ma (U-Pb) volcanic rocks from the Surumu Group, northern Roraima State (Amazonian Craton). After thermal and AF demagnetization, characteristic northwestern remanent magnetization (ChRM) directions with downward inclinations were isolated (mean: $D_m = 298.6°$, $I_m = 39.4°$, $\alpha_{95} = 10.1°$, $N = 20$, $K = 11.4$), which yielded a paleomagnetic pole (SG) located at 234.8°E, 27.4°N ($A_{95} = 9.8°$). Magnetic mineralogy experiments show that ChRM is carried by high-Hc, and high-unblocking temperature Ti-poor titanomagnetite and/or hematite. The well-dated pole SG contributes to better define the apparent polar wander (APW) path for the Guiana Shield during the Paleoproterozoic (2070–1960 Ma). Comparison of the Amazonian and West African Craton's APW paths for the same time interval suggests that these cratonic blocks were linked together at about 1980–1960 Ma ago in a paleogeography along which the Guri (Guiana Shield) and Sassandra (West Africa Craton) shear zones were aligned as previously proposed.

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1. Introduction

Establishing the supercontinental cycle is one of the most important tasks in deciphering the geological evolution of the Earth, particularly for early times when the continental configuration is yet poorly known and intensively debated. The paleogeography of old cratonic blocks is poorly established mainly due to the paucity of high-quality paleomagnetic poles, which makes the Archean-to Paleoproterozoic paleogeographic reconstruction an exercise of strong subjectivity (Pesonen et al., 2003). But several recent contributions have made considerable progress in populating the Paleoproterozoic paleomagnetic database, enabling a first sketch of the 'puzzle' of the Paleoproterozoic Columbia supercontinent assembly and dismissal to be constructed (e.g., Rogers, 1996; Rogers and Santosh, 2002; Zhao et al., 2002, 2003, 2004, 2006; Kusky et al., 2007; Meert, 2002, 2012; Pesonen et al., 2003, 2012; Salminen and Pesonen, 2007; Johansson, 2009; Lubnina et al., 2010; Yakubchuk, 2010; Piper et al., 2011; Evans and Mitchell, 2011; Bispo-Santos et al., 2008, 2012; D'Agrella-Filho et al., 2012; Zhang et al., 2012).

The Amazonian Craton is one of the largest pieces of the Columbia 'puzzle'. It covers almost half the surface of Brazil and parts of Bolivia, Colombia, Venezuela, Guyana, Suriname and French Guiana (Fig. 1) in South America. Its geological evolution is unique compared to that of other cratons in the continent, bearing more similarities with the evolution of Laurentia, Baltica, and West Africa. But notwithstanding its importance, it still exhibits, together with West Africa, one of the poorest paleomagnetic records for the Paleo- and Mesoproterozoic, mainly due to the difficult access and poor outcropping conditions into this densely forested region (Meert and Powell, 2001; Pesonen et al., 2003; D'Agrella-Filho et al., 2012).

Essentially, only two studies have contributed to the Paleoproterozoic paleomagnetic database of the Amazonian craton so far. The first one, from the 1980s, was performed at the northern savannah in Venezuela and Guyana (Guiana Shield) by Princeton's group (Onstott and Hargraves, 1981; Onstott et al., 1984; Renne et al., 1988). These Paleoproterozoic data when compared with those from West Africa Craton suggested that the Guiana Shield was an extension of the West Africa Craton at 2000 Ma ago, but displaced from each other with respect to a Gondwana type resconstruction in such a way that the Paleoproterozoic lineaments Guri (in Guiana Shield) and Sassandra (in West Africa) were aligned (Onstott and Hargraves, 1981). The second study, in the early 2000s, was performed by the BRGM (Bureau de Recherches Géologiques et Minières, France) in granites and metavolcanic rocks from French Guyana (Nomade et al., 2001, 2003). The same group has also obtained data from West Africa (Nomade et al., 2003). Comparing poles from the two cratons, they concluded that the apparent polar

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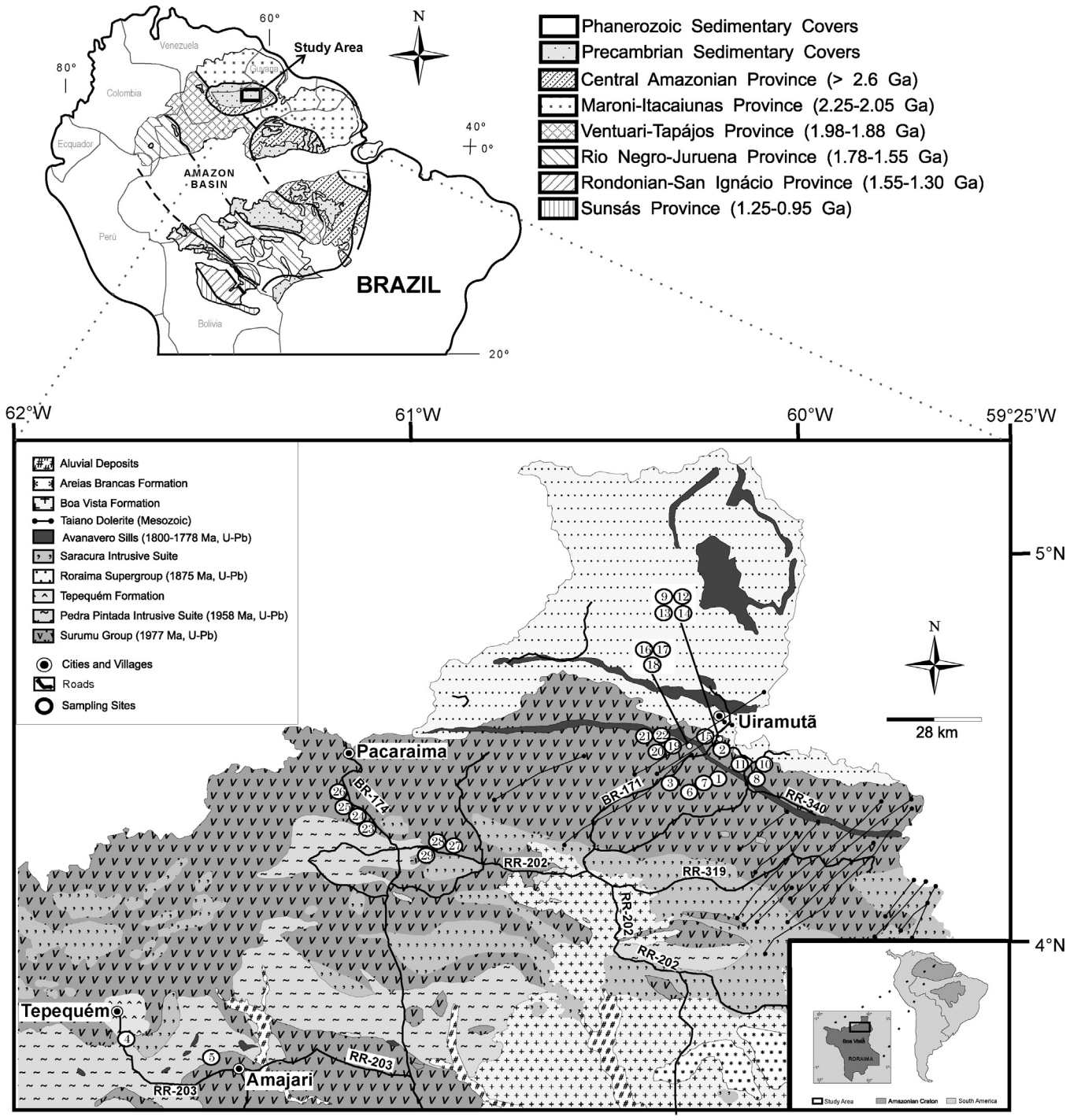


Fig. 1. Geological sketch of the studied area with location of sampling sites (circled numbers) (adapted from Reis et al., 2004). Inset – Amazonian Craton and their geochronological Provinces (after Tassinari et al., 2000).

wander (APW) paths of the Guiana Shield and West Africa Craton crossed over at ca. 2020 Ma and then followed a nearly parallel path for the paleogeographic configuration suggested by Onstott and Hargraves (1981). More recently, Théveniaut et al. (2006) presented an extensive paleomagnetic and geochronological work on Paleoproterozoic plutonic and metamorphic rocks from the French Guiana. Based on their new paleomagnetic poles and reinterpreting previous published poles they proposed a new APW path for Guiana Shield between 2155 Ma and 1970 Ma. Here, we present a paleomagnetic study on well-dated 1980–1960 Ma (U-Pb) volcanic rocks from the Surumu Group that crop out in Roraima State, northern Brazil. The results provide a new robust paleomagnetic pole, which helps to better define the APW path for the Guiana Shield (Amazonian Craton). This new path is compatible with the Onstott and Hargraves (1981) link between the Guiana Shield and West Africa in the Paleoproterozoic at about 2000 Ma ago.

2. Geological setting

The Amazonian Craton (Fig. 1) is exposed in two large areas forming the northern Guiana Shield and the southern Central-Brazil or Guaporé Shield, with the Amazonas sedimentary basin

interposed between these two Precambrian areas (Santos et al., 2000; Lacerda-Filho et al., 2004). Its northeastern part comprises two small Archean nuclei (>2600 Ma, Central Amazonian Province in Fig. 1) whose collision originated the (2250–2050 Ma) Paleoproterozoic Maroni-Itacaiunas Province (Tassinari and Macambira, 1999; Tassinari et al., 2000). Following the assembly of the two Archean nuclei, subduction-related Paleoproterozoic juvenile magmatic arcs were formed at 1980–1810 Ma (Ventuari Tapajós) and 1780–1550 Ma (Rio Negro–Juruena) (Tassinari et al., 2000; Pinho et al., 2003; Schobbenhaus and Brito-Neves, 2003; Cordani and Teixeira, 2007). In the Mesoproterozoic, subduction-related magmatic arcs in Mato Grosso and Rondônia States culminated with the collision of the Paraguá Terrane at ca. 1300 Ma forming the 1600–1300 Ma Rondonian-San Ignacio Province (Bettencourt et al., 2010; Rizzotto and Hartmann, 2012). The southwestern tip of the Amazonian Craton comprises the 1200–950 Ma Sunsás-Aguapeí Province, which is considered the counterpart of the Greenville Province in Laurentia (Teixeira et al., 2010).

The study area is located in the northern Roraima State, inserted into the Central Amazonian geochronologic Province (Fig. 1). The Central Amazonian Province (CAP) is composed of Archean continental crust (>2600 Ma) which was covered by a large Paleoproterozoic volcano-sedimentary succession: the 1980–1960 Ma Surumu acid volcanic rocks and the (ca. 1870 Ma) sedimentary sequences of the Roraima Supergroup (Teixeira et al., 1989; Tassinari and Macambira, 1999; Santos et al., 2003; Reis et al., 2003). As a whole, the northern portion of Roraima State is represented by a volcano-plutonic and intracratonic sedimentary terrain where the main Roraima basin has been installed. The U-Pb SHRIMP ages in the Brazilian portion show an interval of 1980–1950 Ma to its evolution. This area prolongs to Venezuela, Guyana and Suriname. The volcano-plutonism was established around 1980 Ma (U-Pb SHRIMP) and is represented by the Surumu Group (1982 ± 3 Ma), Pedra Pintada (1985 ± 1 Ma) and Aricamã (1986 ± 4 Ma) suites (Fraga et al., 2009; Fraga and Dreher, 2010). The volcano-plutonic association shows post-collisional characteristics and also includes A-type rocks.

In the Surumu–Cotingo area (this study) the Surumu volcanics and Pedra Pintada Suite were grouped into the Orocaima magmatic arc (Reis et al., 2000) with younger U-Pb ages at 1966 ± 9 Ma (Schobbenhaus et al., 1994) and 1956 ± 5 Ma (Santos, 2003). It corresponds to a new accretionary stage and suggests a collisional environment younger than the Trairão–Anauá arc (2030 Ma, Fraga and Dreher, 2010). Granitoid bodies with late- to post-collisional characteristics have been related to the Saracura Suite with Rb–Sr ages at 1800–1700 Ma (Fraga and Araújo, 1999; Menezes Leal et al., 2006).

The Roraima basin shows thermal and mechanical processes of subsidence and was developed as a pull-apart type according to its rhombohedral relic pattern and depositional environments with source area to the northeast (Reis et al., 1990). The Avanavero magmatism represented by mafic dykes and sills establishes the minimum U-Pb age in 1782 ± 3 Ma (Santos et al., 2003; Reis et al., 2013) to the Roraima Supergroup.

3. Sampling and analytical methods

In northern Roraima State, 225 oriented cylindrical cores and two oriented block samples were collected from 39 well-exposed sites of acid to intermediate volcanic rocks from the Surumu Group close to the Amajari and Pacaraima towns, the Uiramutã and Surumu villages and the Tepequém Mountain (Fig. 1). Along the Cotingo River, close to the Uiramutã village, a Mesozoic N45E mafic dyke cuts the Surumu rocks. Only one of the contacts was visible but the exposed surface shows that the dyke's width is surely larger than 20 m. At this location a detailed sampling was performed to verify the magnetic stability of the Surumu volcanics and the Mesozoic dyke. Four cylindrical cores were sampled from the dyke (site 15: FR95–cores A–D in Table 1) and 8 cores were extracted from the host acid rock (Surumu Group) progressively away from the dyke's border, from 1.70 m to 21.70 m (site 15: FR95–cores E–L). The visible contact with the host rock was under water, preventing sampling directly at the contact. In addition, 18 cores were sampled from three other Surumu sites at greater distances from the contact with the dyke: site 16 (FR96 – cores A–F) located ~33 m from the contact, site 17 (FR 97 – cores A–F) located ~70 m from the contact, and site 18 (FR98 – cores A–F) located ~100 m from the contact.

Samples were oriented using both sun and magnetic compasses. At the laboratory the cylindrical cores were cut into 2.2 cm height specimens. Step-wise alternating magnetic field (AF) and thermal demagnetization techniques were employed in 445 specimens to separate the characteristic remanent magnetization (ChRM) component carried by the samples. Normally, steps of 2.5 mT (up to 15 mT) and 5 mT (15–100 mT) were employed for AF demagnetization using a tumbler Molspin AF demagnetizer. Thermal demagnetization was performed using a Magnetic Measurements TD-60 furnace in steps of 50 °C (from 150 °C up to 500 °C) and 20 °C (from 500 °C up to 600 °C/700 °C). Remanent magnetization measurements were carried out using a JR-6A spinner magnetometer (AGICO, Czech Republic). These instruments are housed in a magnetically-shielded room with ambient field < 1000 nT. Magnetic components for each specimen were identified in orthogonal plots (Zijderveld, 1967), and calculated using principal component analysis (Kirschvink, 1980). At least 4 demagnetization steps were used to calculate vectors using least-squares fits, and an upper limit for mean angular deviation (MAD) of 8° was used. Fisher's (1953) statistics were used to calculate site mean directions and the paleomagnetic pole.

Magnetic mineralogy was investigated through acquisition of isothermal remanent magnetization (IRM) curves using a pulse magnetizer (Magnetic Measurements), and hysteresis curves using a MicroMag VSM – Model 3900 (Princeton Measurements Corporation). Hysteresis curves give bulk coercive force (H_c), saturation magnetization (M_s), and saturation isothermal remanence (M_{rs}), after subtraction of the paramagnetic contribution. Coercivity of remanence (H_{cr}) was obtained using the same Micromag VSM instrument. Also, thermomagnetic curves (low-field magnetic susceptibility versus temperature from −194 to 700 °C) were performed for several samples to characterize the magnetic carriers in the rock using a CS-4 apparatus coupled to the KLY-4S Kappabridge instrument (AGICO, Czech Republic). Thin and polished sections were microscopically investigated under transmitted and reflected light, respectively.

4. Paleomagnetic results

A summary of paleomagnetic results is shown in Table 1. AF demagnetization was efficient in isolating the characteristic remanent magnetization (ChRM) for most samples, although hematite present in some samples was only demagnetized after thermal treatment. Secondary components were eliminated on alternating fields up to 15 mT and at temperatures below 500–540 °C. At higher AF fields and temperatures, a northwestern ChRM component with downward/upward inclination (Fig. 2) was isolated for 207 specimens from 22 sites (Table 1), which was named Surumu Group (SG) component. Samples belonging to 7 other sites (4, 5, 13, 15, 23, 26 and 27) show coherent within-site magnetization directions but different from the SG component (Table 1). Three of them (sites 5, 13 and 15 in Table 1) revealed northern directions with low downward inclinations very close to those obtained for

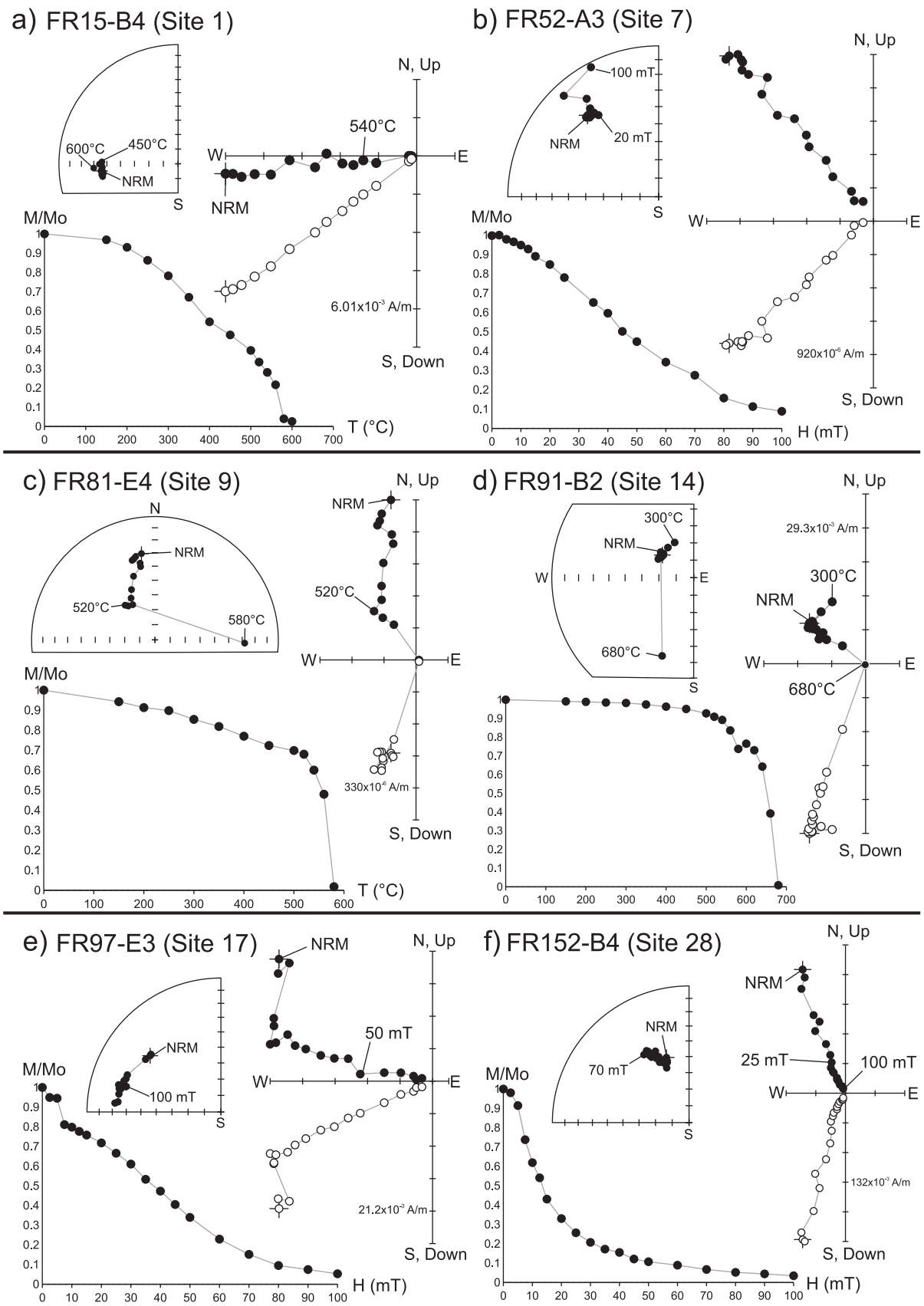


Fig. 2. Examples of magnetic behavior of six samples from different sites submitted to AF or thermal demagnetizations. Figure shows stereographic projections (solid symbols represent positive inclinations), normalized intensity curves (M/M_o versus alternating field (H)) and orthogonal projections (solid (open) symbols represent horizontal (vertical) projections) for each sample.

Table 1
Paleomagnetic data from Surumu Group volcanic rocks.

Site	Localization	Samples	Site mean direction					VGP	
			N/*n*	D_m (°)	I_m (°)	α_{95} (°)	*k*	Plong (°E)	Plat (°N)
1	04.53°N/299.80°E	FR15A-E	12/10	277.5	44.3	11.7	18.1	235.5	8.7
2	04.54°N/299.82°E	FR18A-F	16/11	298.1	62.3	13.9	11.8	255.8	23.2
3	04.49°N/299.74°E	FR19A-F	13/11	320.4	28.0	12.7	13.9	227.5	49.7
4	03.74°N/298.27°E	FR22A-F[a]	10/7	240.4	25.2	13.6	20.7	–	–
5	03.74°N/298.47°E	FR24A-F[a]	10/9	19.4	4.3	9.8	28.8	–	–
6	04.48°N/299.76°E	FR50A-F	12/12	335.8	13.6	6.8	41.5	216.6	65.8
7	04.47°N/299.83°E	FR52A-F	12/12	311.7	36.4	3.2	189.0	232.7	40.5
8	04.43°N/299.86°E	FR80A-D[a]	12/11	297.7	−12.5	7.7	36.5	20.5	−26.9
9	04.52°N/299.79°E	FR81A-G	12/10	336.9	38.8	11.5	18.6	249.7	61.7
10	04.44°N/299.89°E	FR82A-F	12/12	282.0	37.7	9.6	21.4	230.6	12.8
11	04.43°N/299.87°E	FR83A-F	12/7	299.0	8.0	10.9	31.8	212.0	29.2
12	04.52°N/299.79°E	FR89A-F	12/7	311.2	47.3	14.7	17.7	242.8	37.9
13	04.52°N/299.79°E	FR90A-F[a]	6/6	12.9	1.1	4.2	256.8	–	–
14	04.52°N/299.79°E	FR91A-D	12/12	282.8	67.6	4.2	106.5	260.5	11.6
15[b]	04.52°N/299.73°E	FR95A-D[a]	5/5	17.1	5.0	6.2	155.7	–	–
		FR95E-L[a]	13/10	9.0	14.6	4.4	118.9	–	–
16	04.49°N/299.73°E	FR96A-F	14/12	273.5	11.3	4.8	83.7	215.2	3.9
17	04.49°N/299.73°E	FR97A-F	12/12	278.3	21.0	3.3	175.2	220.1	9.0
18	04.49°N/299.73°E	FR98A-F	14/14	305.7	23.7	7.1	32.7	221.8	35.8
19	04.50°N/299.67°E	FR99A-H	10/5	291.0	17.9	23.4	11.7	217.8	21.4
20	04.50°N/299.67°E	FR100A-F	12/8	295.5	44.3	11.1	25.9	236.3	24.8
21	04.51°N/299.67°E	FR102A-F	12/12	259.1	66.7	9.6	21.6	259.7	−3.6
22	04.51°N/299.67°E	FR103A-F	12/8	301.5	37.9	14.9	14.8	231.8	30.9
23	04.28°N/298.93°E	FR146A-F[a]	8/6	154.5	24.8	14.5	22.2	–	–
24	04.29°N/298.93°E	FR147A-F	11/11	310.9	54.9	9.6	23.7	250.1	35.1
25	04.31°N/298.92°E	FR148A-E[a]	13/12	327.9	−3.0	8.0	30.1	19.3	−57.4
26	04.33°N/298.88°E	FR149A-E[a]	10/5	346.6	−75.0	10.1	58.8	–	–
27	04.21°N/299.15°E	FR151A-E[a]	10/5	106.3	59.1	17.3	20.6	–	–
28	04.21°N/299.12°E	FR152A-E	13/10	301.7	51.3	11.5	18.6	246.3	36.2
29	04.22°N/299.08°E	FR153A-E	13/11	268.3	38.2	6.8	46.5	230.6	0.0
Mean			**20**	**298.6**	**39.4**	**10.1**	**11.4**		
SG pole						**9.8**	**12.1**	**234.8**	**27.4**

Footnote: *n*/*N* – number of analyzed specimens/number of specimens used in the mean; D_m – Declination; I_m – Inclination; α_{95}, k – Fisher's confidence cone and precision parameter (Fisher, 1953); VPG – Virtual Geomagnetic Pole; Plong. – Paleolongitude; Plat – Paleolatitude.

[a] Excluded sites from the mean.

[b] Site 15: cores FR95A–FR95D are from the Mesozoic dyke; cores FR95E–FR95L are from the Surumu rocks collected up to 21.70 m from the contact.

Mesozoic dykes (Théveniaut et al., 2006), which are common in the studied area (Reis et al., 2013). In particular, samples from site 15 (samples FR95E–FR95L) were remagnetized by a Mesozoic dyke (see below). Samples belonging to site 23 (Table 1) yielded southeastern directions with low downward inclinations (D_m = 154.5°, I_m = 24.8°), similar to the directions obtained for the ~1790 Ma Avanavero sills (mean: D_m = 138.2°, I_m = −3.4°; *N* = 13; α_{95} = 13.0°), which cut across the Surumu rocks (Bispo-Santos et al., 2014, in this issue). The remaining three sites (4, 26 and 27, Table 1) presented very different directions which do not match any known magnetic component previously obtained in the region. The origin of these components and their ages are unconstrained. Finally, the other 10 analyzed sites from Surumu Group (not shown in Table 1) presented magnetic instability and/or within-site magnetic inconsistency.

Sites 8 and 25 showed biotite rims in martite or hematite crystals, suggesting chemical alteration (see below). These two sites are the ones with larger scatter relative to the other sites, and were thus excluded from the global mean. Site mean directions for the remaining 20 sites (Fig. 3) cluster around the mean at: D_m = 298.6°; I_m = 39.4° (*N* = 20, α_{95} = 10.1°, *K* = 11.4), yielding a paleomagnetic pole (SG) located at 234.8°E, 27.4°N (A_{95} = 9.8°, *K* = 12.1) with angular dispersion of *s* = 23.6°.

5. Magnetic stability test

A baked contact test was performed at the Cotingo River where a Mesozoic dyke cuts across Surumu volcanic rocks (see sampling description in topic 3 above). Dyke's samples revealed north/northeastern directions with low downward inclinations (Fig. 4a), typical of Mesozoic dykes cropping out in northern Amazonian Craton (e.g., Veldkamp et al., 1971; Théveniaut et al., 2006). Similar directions were disclosed for volcanic rocks collected up to 21.7 m away from dyke's contact. Fig. 4b shows an example given by site 15 (sample FR95-J1), situated 15 m away from the contact. In contrast, samples belonging to sites 16 (33 m from the contact), 17 (70 m from the contact) and 18 (100 m from the contact) revealed west/northwestern directions with downward inclinations, typical of the Surumu component (Fig. 4c and d, Table 1). This baked contact test demonstrates the magnetic stability of the Mesozoic

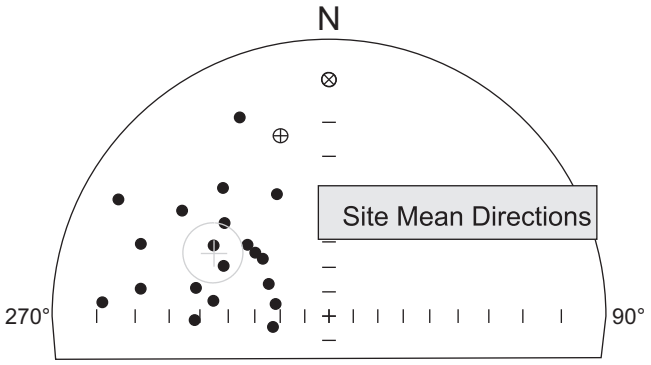


Fig. 3. Site mean directions for the Surumu Group rocks. Solid circles represent downward inclinations. Plus signal represents the mean of site mean directions with its respective confidence circle (α_{95} = 10.1°). ⊗ – present dipolar geomagnetic field; ⊕ – present geomagnetic field.

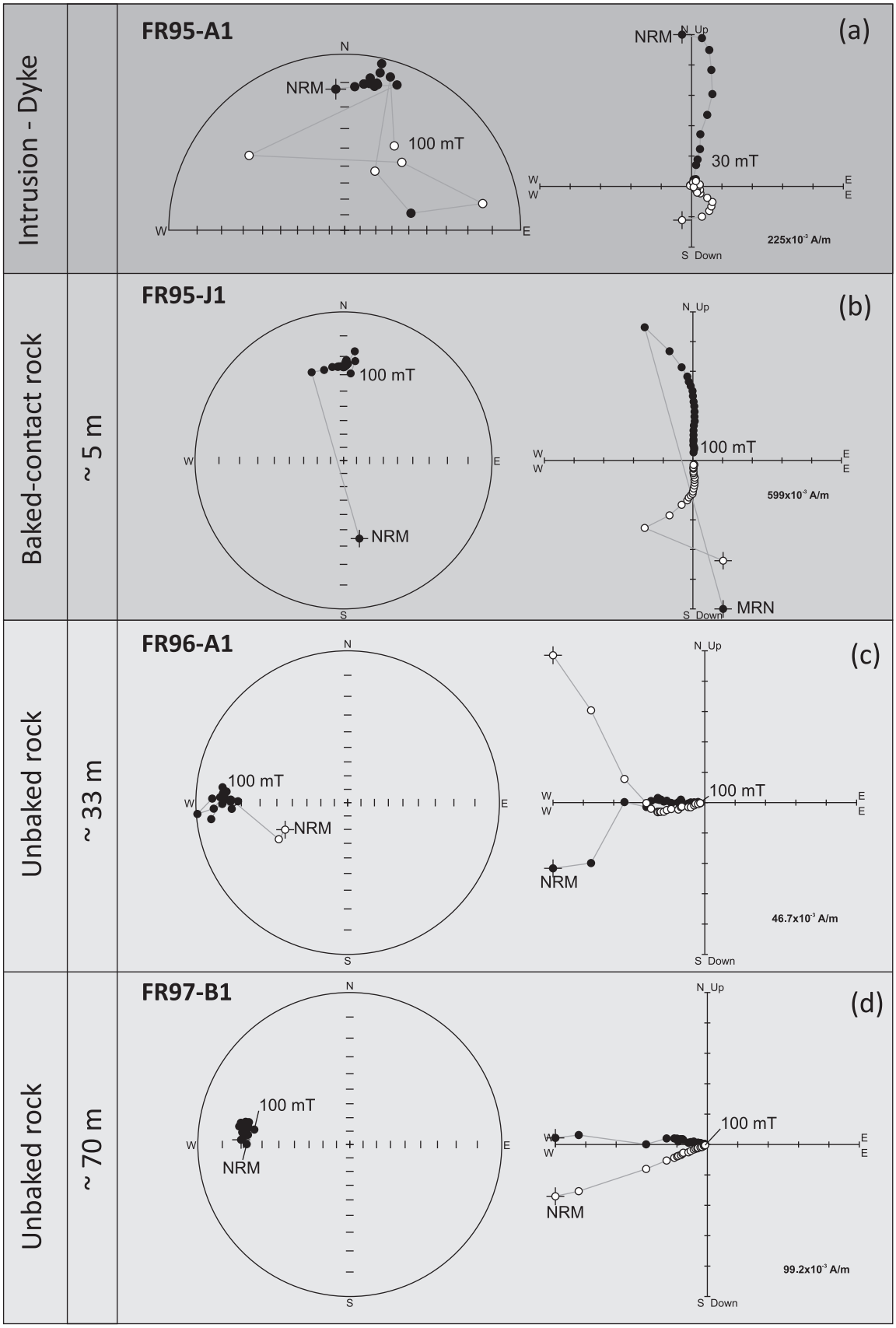


Fig. 4. Baked contact test: AF demagnetization of a sample from the Mesozoic mafic dyke (a), of a Surumu baked rock sample at ~15 m from the intrusion (b), and of the Surumu unbaked rock samples at ~33 m (c) and at ~70 m (d) from the contact. Figure shows stereographic projections (solid (open) symbols represent positive (negative) inclinations), orthogonal projections (solid (open) symbols represent horizontal (vertical) projections), and normalized intensity curves (M/M_o versus alternating field (H)) for each sample.

dyke and its primary origin. At the same time, it shows that the Surumu component was not affected by recent remagnetizations. This inverse baked contact test and the fact that the Surumu component is quite different from the other magnetic components disclosed so far for younger Amazonian craton units, including that of the 1788.5 ± 2.5 Ma Avanavero sills, for which a positive baked contact test was obtained (Bispo-Santos et al., 2014, in this issue), suggests it may represent a primary magnetization acquired during cooling of the Surumu volcanics at about 1980–1960 Ma.

6. Magnetic mineralogy

Fig. 5a shows typical examples of normalized NRM intensity curves after AF demagnetization. Most samples show coercivities below 100 mT, typical of titanomagnetite. Low coercivity fractions (<10–15 mT) are associated with these samples, which normally carry secondary components (see description above). Higher coercivity fractions up to 30–40 mT (e.g., for samples FR82-C2, FR100-C3 and FR152-C1, Fig. 5a) or even greater than 100 mT (e.g., FR19-B2 and FR96-A1, Fig. 5a) are associated to the ChRM directions isolated in the volcanic rocks. For some samples, however, the main magnetic carrier is probably hematite, as evidenced by the low decay of magnetization intensity (less than 20%) in AF fields as high as 100 mT (e.g., sample FR91-D1/site 14 in Fig. 5a)

Fig. 5b shows typical examples of normalized intensity curves after thermal demagnetization. Most samples yielded distributed unblocking-temperature (T_{ub}) spectra, which visibly show two T_{ub} intervals (samples FR15-C2, FR19-A2 and FR81-E4 in Fig. 5b): a lower-T peak (<500–550 °C), normally associated with secondary components, and a higher-T peak (500/550–580 °C), indicative of magnetite as the main carrier of ChRM directions. Sample FR91-C3 (site 14) yielded mixed magnetic mineralogy with T_{ub} as high as 680 °C indicating the additional presence of hematite (Fig. 5b). Particularly, this sample shows an intensity decay of only 20% between 550 °C and 580 °C and a second decay between 640 °C and 680 °C; a sister sample (FR91-D1) presented a similar decay in intensity at fields below 100 mT (Fig. 5a).

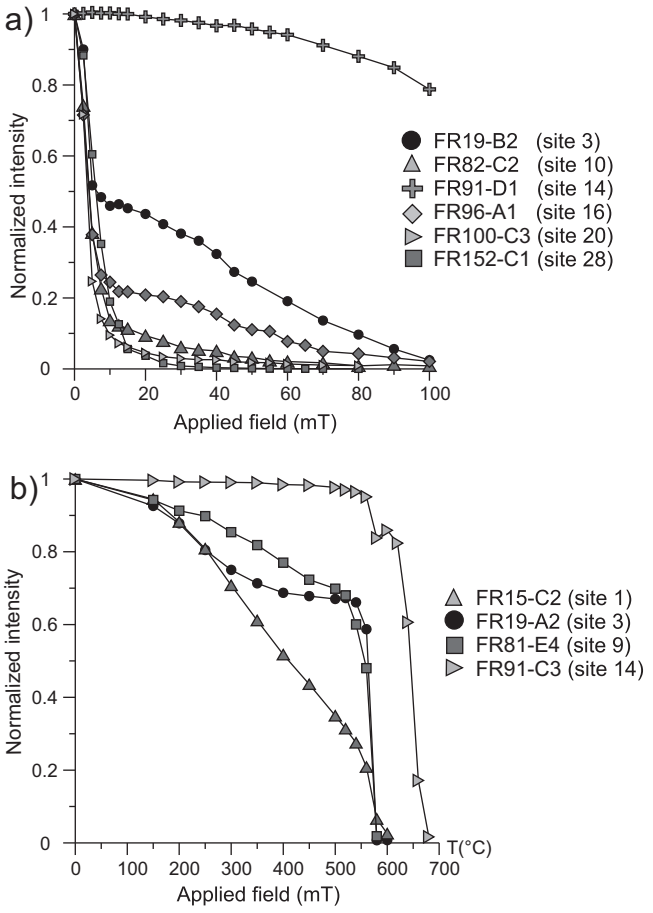


Fig. 5. Examples of normalized magnetization intensities after (a) alternating magnetic field and (b) thermal demagnetization for samples from different sites.

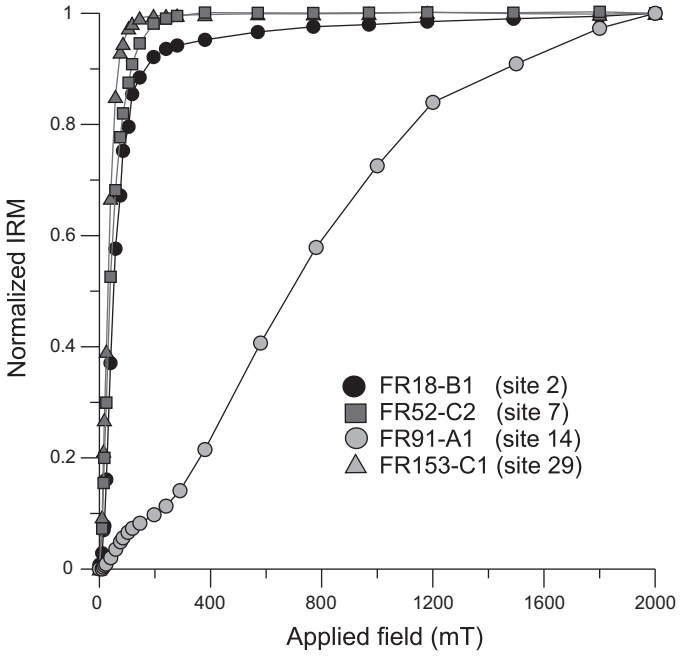


Fig. 6. Isothermal remanent magnetization (IRM) acquisition curves (normalized intensity versus magnetic field).

Isothermal remanent magnetization (IRM) curves obtained for most samples showed a pronounced increase in magnetization in fields below 300 mT (Fig. 6) emphasizing the presence of titanomagnetite/magnetite. However, some samples only reach saturation in magnetization at fields greater than 1800 mT (Fig. 6). Such behavior corroborates the presence of hematite in these rocks as evidenced by AF and thermal treatments. Again, a sample (FR91-A1) from site 14 presents a distinct behavior, where IRM intensity, after a rapid increase up to 150–200 mT (associated with the presence of magnetite), gradually reaches saturation in fields higher than 2000 mT (associated with hematite).

Narrow-waisted hysteresis curves giving low coercivities (2.7–11.2 mT), and typical of titanomagnetite, were obtained for many samples. The presence of hematite observed in demagnetization patterns and IRM curves is also observed in the hysteresis curve obtained for sample FR91-B (site 14), which shows a larger coercivity (H_c = 40.6 mT, Fig. 7). Fig. 8 shows the Day's diagram (Day et al., 1977, modified by Dunlop, 2002) indicating that most samples fall into the pseudo-single domain (PSD) field or along a trend parallel to the theoretical SD/MD mixing curves of Dunlop (2002). This behavior is consistent with the good magnetic stability obtained during AF and thermal treatments. However, some samples exhibit magnetic grains into the multi-domain (MD) field. This behavior can explain the large portion of low coercivity remanence and magnetic instability in many specimens.

Both reversible and irreversible high-temperature thermomagnetic curves were obtained for the studied rocks, most of them with a pronounced Hopkinson peak and a strong decay at around 580 °C (Fig. 9). In all irreversible curves (e.g., Figs. 9a and b) susceptibility increases during the cooling phase, indicating

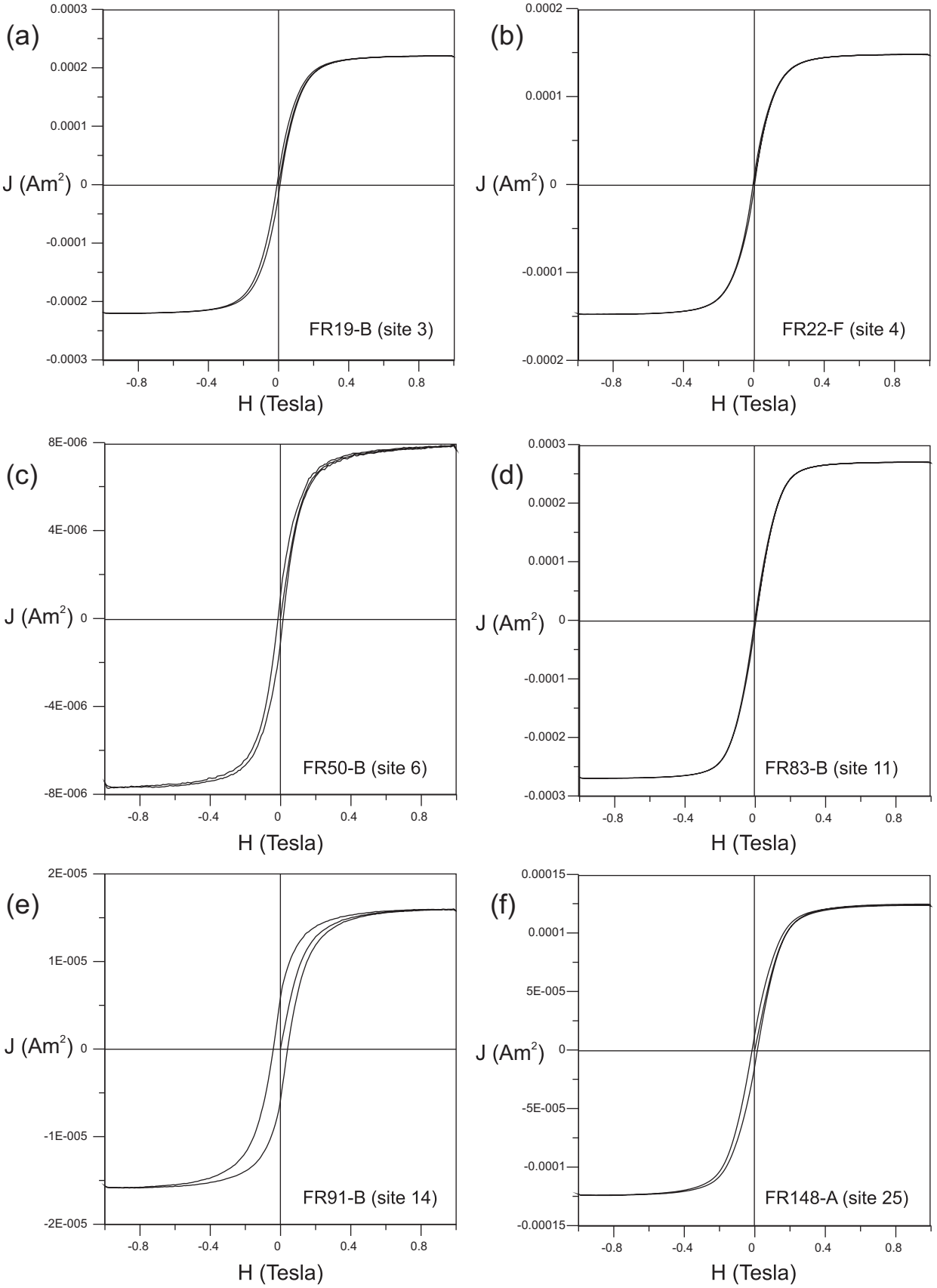


Fig. 7. Narrow waist type hysteresis curves (magnetic moment (*J*) versus magnetic field (*H*)) characteristic of titanomagnetite/magnetite.

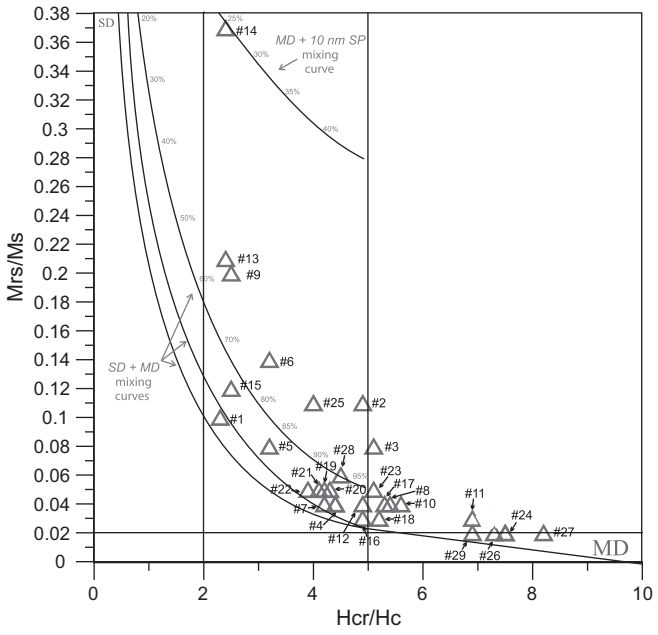


Fig. 8. Day's diagram (Day et al., 1977) after Dunlop (2002): M_{rs}/M_s versus H_{cr}/H_c ratios. Most samples fall in the Pseudo-single domain (PSD field), or along the SD (Single domain) plus MD (Multidomain) mixing curves as proposed by Dunlop (2002). Percentages of MD grains in the mixture are also shown. Numbers (#) indicate the analyzed sites.

that probably magnetite is being formed during heating. Low-temperature thermomagnetic curves show a susceptibility peak at around −153 °C. These characteristics are typical of SD/PSD magnetite grains (Dunlop and Ozdemir, 1997), which is probably the main magnetic carrier in these rocks. For many samples, however, the curves suggest the additional presence of hematite as indicated by the fall of magnetic susceptibility at temperatures above 600 °C (Fig. 9). In particular, sample FR91-B (site 14) shows a small contribution of magnetite with predominance of hematite, as already shown by the other magnetic experiments (see above).

7. Petrography

The analyzed sites of the Surumu Group are composed of rhyolite with feldspar megacrystals in micropoikilitic matrix, ignimbrites with feldspar, and volcanic glass. Some of the samples show devitrification structures (spherulites). Glass shards deformed around megacrystals in ignimbrite (Plate 1A), as well as perlitic texture in volcanic glass (Plate 1B), indicate a high temperature for formation of these lithotypes (Fisher and Schmincke, 1984; McPhie et al., 1993). Both rhyolite and pyroclastic rocks show feldspar megacrysts, sometimes preserving original flow orientation. Quartz and chlorite filling in amygdales are common. Well-preserved magnetite is the dominant observable magnetic phase (Plate 1C). Partial alteration to martite is present in up to 10% of the magnetite crystals. Titanomagnetite is found in some sites, such as site 28 (FR152) (Plate 1D), with subordinated occurrence of martite and hematite. Chrystal-rich ignimbrite shows magnetite in amygdales oriented according to the flow direction (Plate 1E). Basalts occur locally (sites 12: FR89 and 14: FR91), showing trachytic texture, partially chloritized matrix and amygdales filled with quartz, martite and, locally, magnetite (Plate 1F). Martite with partial dissolution textures is found in the matrix of those rocks (Plates 1G1/1G2), suggesting hydrothermal activity, possibly related to the later Surumu volcanism, remagnetizing the basic lithotypes. These petrographic observations are corroborated by the magnetic experiments performed in samples from site 14 (FR91), which revealed hematite as the predominant magnetic carrier in these rocks (see Section 6).

Surumu samples showing a different magnetic component or unstable behavior are mostly rhyolites very often sericitized, in micropoikilitic matrix. Observable magnetic minerals are mainly hematite and martite (Plate 2A). Partially preserved magnetite crystals are also present, sometimes replaced by titanite (Plate 2B). Remagnetization in these rocks is suggested by the ubiquitous presence of recrystalization textures, including quartz and feldspar crystals larger than those in the matrix (Plate 2C) or replacing other minerals (Plate 2D). In some sites biotite rims in martite or hematite crystals (Plate 2E–sites 8: FR80-C1; 23: FR146-D1; 25: FR148-B3; 26: FR149-B1; 27: FR151-D1).

8. Discussion

8.1. Reliability of the Surumu paleomagnetic pole

Stable northwestern, downward inclination directions (SG component) were determined for 207 specimens from 20 out of 39 analyzed sites of the Surumu volcanic rocks. The SG component is carried by PSD magnetite and/or hematite with high-coercivities and high-T_{ub}, attesting to its high magnetic stability. An inverse positive baked contact test was obtained for a Mesozoic dyke cutting across the volcanic rocks. Although this test did not prove the primary nature of the SG component, it indicates that no recent remagnetization occurred in those rocks. Also, the ca. 1788 Ma Avanavero sills that cut across the Surumu volcanic rocks yielded a very different ChRM direction ($D_m = 138.2°$, $I_m = -3.4°$, $N = 13$, $\alpha_{95} = 13.0°$ – paleopole at 27.9°E, 48.4°S, $A_{95} = 9.6°$), for which a positive baked contact test was obtained (for details see Bispo-Santos et al., 2014, in this issue). From the above the SG component most probably represents a primary magnetization acquired during volcanic rock cooling.

The Surumu Group (SG) pole (27.4°N, 234.8°E, $A_{95} = 9.8°$) satisfies 5 out of the 7 quality criteria proposed by Van der Voo (1990): (1) the age of the Surumu volcanic rocks can be considered well-determined by U-Pb and $^{207}Pb/^{206}Pb$ methods (Schobbenhaus et al., 1994; Santos et al., 2003; Dreher and Fraga, 2010), with ages between 1980 Ma and 1960 Ma, which may correspond to the time of acquisition of the characteristic remanent magnetization (SG component); (2) the SG pole was derived from a set of 207 specimens (20 sites), and has adequate Fisher's statistical parameters ($A_{95} = 9.8°$, $K = 12.1$); (3) remanence vectors were isolated by least-squares fit from orthogonal diagrams (Kirschvink, 1980) after stepwise AF and thermal demagnetization; (4) the Surumu rocks in the investigated area are situated in the Archean Amazonia-Central Province, which remained stable during the evolution of the Ventuari-Tapajós (1980–1810 Ma) and Rio Negro-Juruena (1780–1550 Ma) provinces, although the mismatch between coeval 1780 Ma paleomagnetic poles from Avanavero sills (Northern Amazonian Craton) and Colider Suite (Southern Amazonian Craton) suggests relative movements within the Amazonian Craton (Reis et al., 2013). Also, the analyzed rocks are spread over a large area, with sampling sites showing coherent directions for distances greater than 200 km (from northwest of Uiramutã village to the Tepequém Mountain; Fig. 1), which means that most probably no tectonic tilting affected this large area; (7) paleomagnetic directions isolated for the Surumu Group are different from those obtained for younger rocks in the study area (Avanavero sills and Mesozoic dykes). The two criteria not satisfied are: (5) only a single polarity was found for the investigated samples and (6) although the inverse magnetic stability test shows that the Surumu ChRM directions (SG component) are older than those found for Mesozoic dykes in the region, it does not prove the primary origin of the

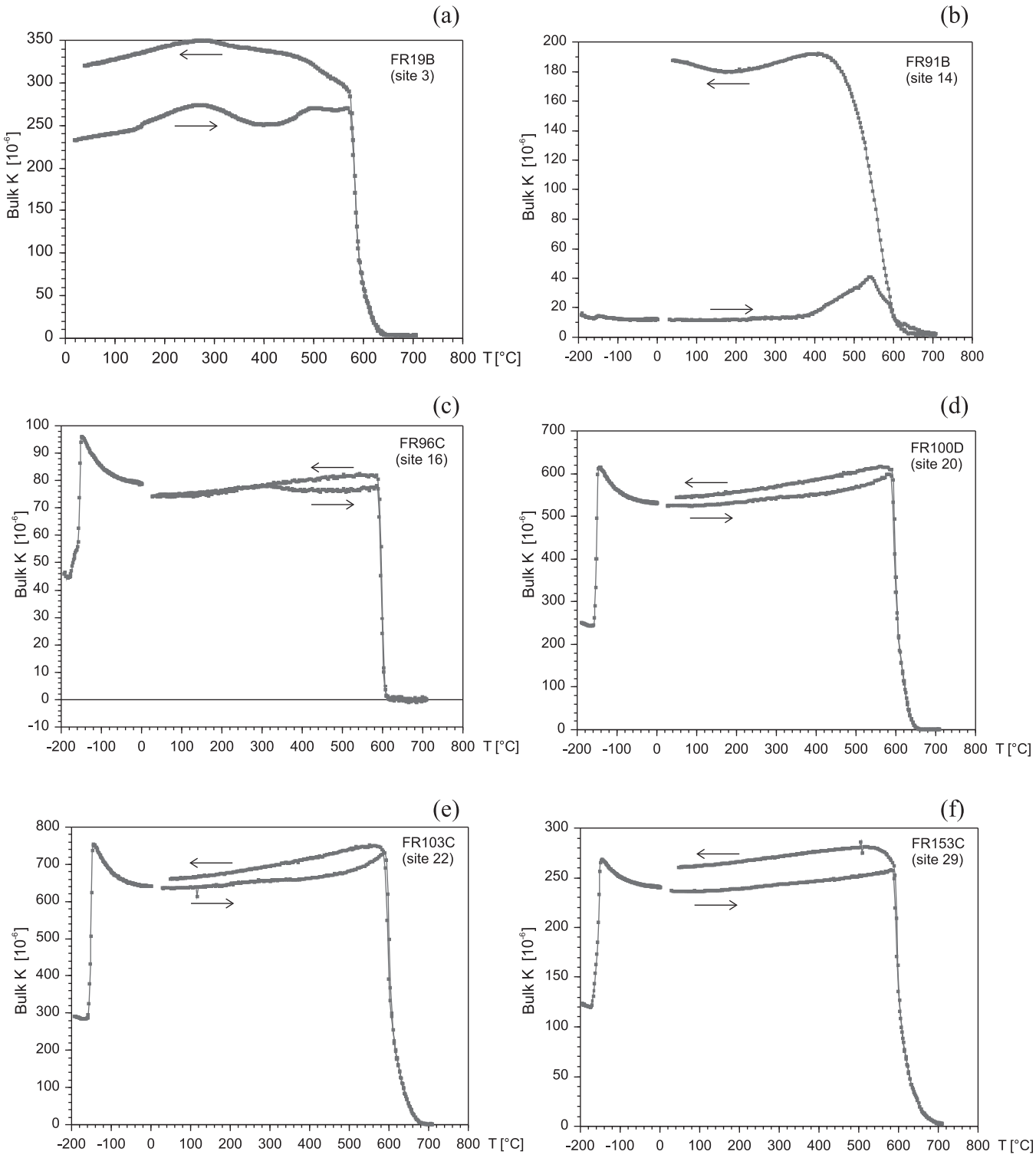


Fig. 9. Thermomagnetic curves showing variation in magnetic susceptibility *K*(SI) versus low and high temperature. Curves were corrected from furnace effects.

SG component. However, as discussed above, the significantly different ChRM direction of Avanavero sills (Bispo-Santos et al., 2014, in this issue) suggests that the SG component is at least older than 1788 Ma.

8.2. The Paleoproterozoic APW path for the Guiana Shield

Two Paleoproterozoic apparent polar wander (APW) paths have been proposed for the Guiana Shield (Amazonian Craton). The first one was proposed by Nomade et al. (2003) for the time interval between 2040 and 1990 Ma. It was based on two paleomagnetic poles obtained by these authors on granitic and metamafic rocks – named as GUI1 (2014 ± 27 Ma) and GUI2 (1993 ± 25 Ma)– and on the selection of paleomagnetic poles obtained in previous studies in the Guiana Shield (Onstott and Hargraves, 1981; Onstott et al., 1984; Nomade et al., 2001). Among the seven poles used to define this APW path, only three were considered by the authors as key poles: the GUI1 and GUI2 poles (Nomade et al., 2003) and the

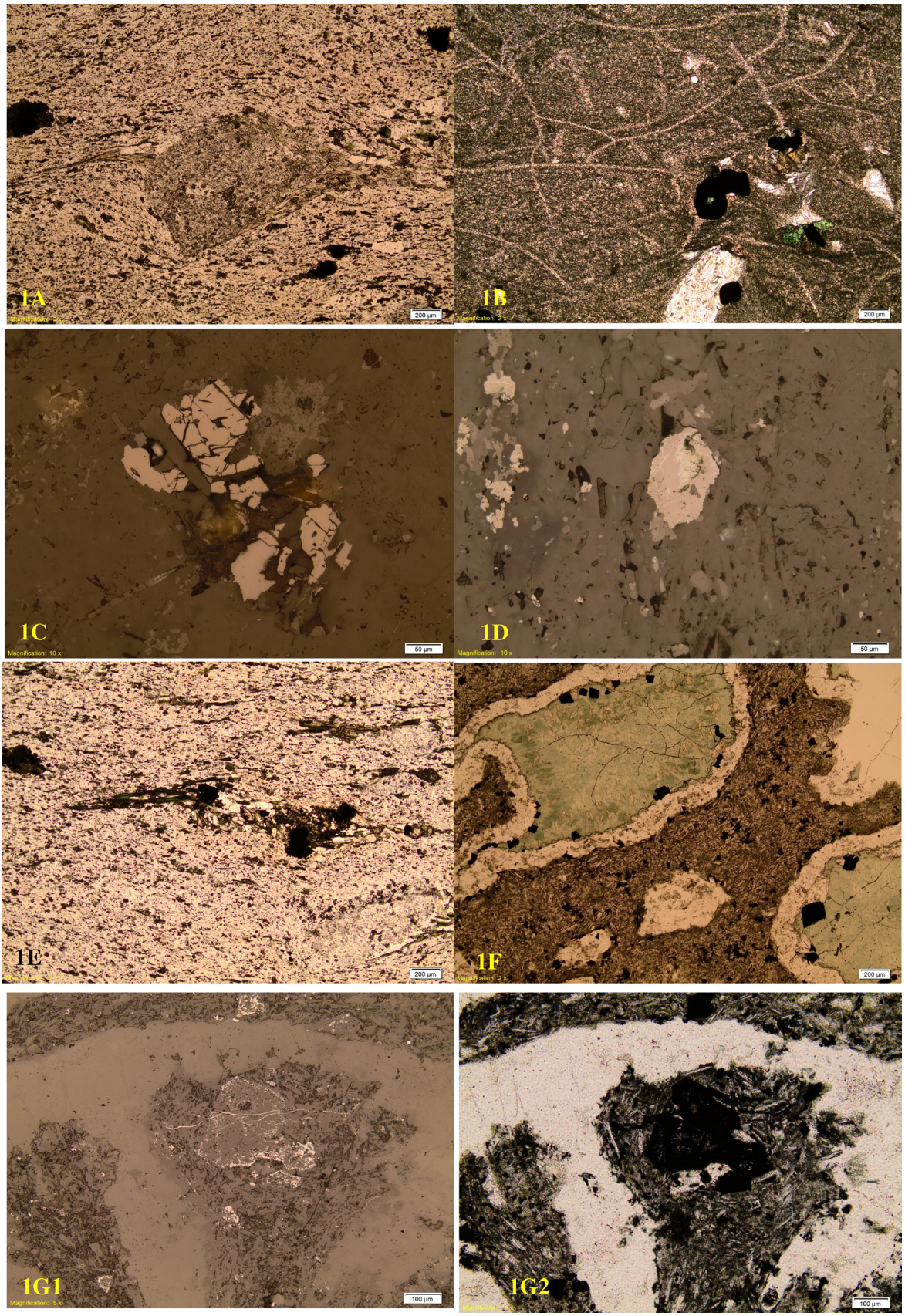


Plate 1. Photographs of samples with SG component – (A) Glass shards deformed around feldspar megacrystal (site 28: FR152 – transmitted light, parallel polarizers); (B) perlitic texture suggesting high temperatures of formation of the pyroclastic rocks (site 3: FR19 – transmitted light, parallel polarizers); (C) preserved magnetite, and (D) titanomagnetite (sites 11: FR83 and 28: FR152, respectively – reflected light, parallel polarizers); (E) titanomagnetite in amygdales oriented according to pyroclastic flow (site 28: FR152 – transmitted light, parallel polarizers); (F) basalt with trachytic texture, partially chloritized matrix and amygdales filled with quartz, chlorite and magnetite (site 12: FR89 – transmitted light, parallel polarizers); (G1/G2) dissolution features in oxide, possibly due to hydrothermal solution related to the Surumu Group, affecting an older basalt (site 14: FR91 – parallel polarizers – reflected (G1) and transmitted (G2) lights, respectively).

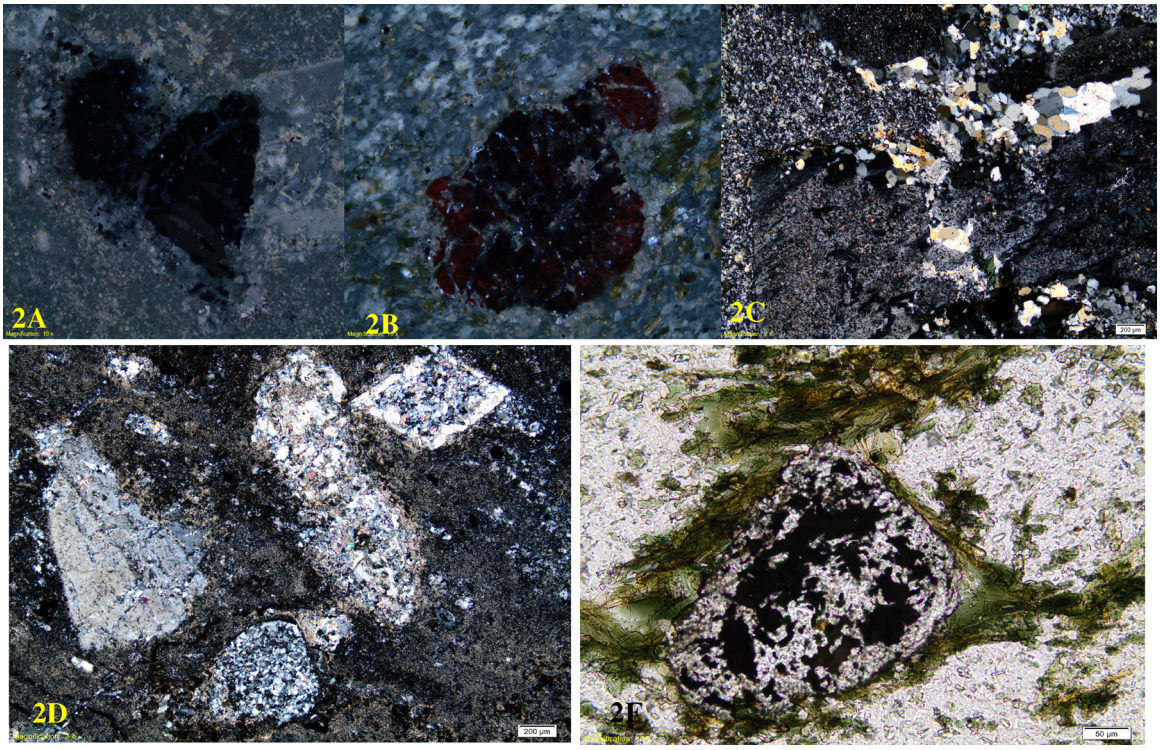


Plate 2. Photographs of samples with other directions–(A) martite and (B) magnetite partially replaced by rutile (sites 4: FR22 and 25: FR148, respectively, transmitted light, parallel polarizers); (C) apparent recrystalization textures with quartz (dominant) and feldspar showing larger crystals than those in the matrix, or (D) replacing other minerals (sites 23: FR146 and 4: FR22, respectively – transmitted light, parallel polarizers): (E) magnetite partially replaced by martite or translucent mineral (titanite?), and involved by biotite and chlorite (site 25: FR148, transmitted light, parallel polarizers).

2036 ± 14 Ma (biotite ^{40}Ar–^{39}Ar age) OYA pole obtained on tonalite and meta-ultrabasite rocks (Nomade et al., 2001). The other less reliable four paleopoles were determined by Onstott et al. (1984) and Onstott and Hargraves (1981), and were used as accessory poles to trace this APW path.

The second APW path was proposed by Théveniaut et al. (2006). In their work, they presented new paleomagnetic data obtained for metamorphic and plutonic rocks from French Guiana, related to the Paleoproterozoic event that formed the Maroni-Itacaiunas Province (e.g., Tassinari et al., 2000). Based on these new data and reinterpreting the ages associated with previous published poles they proposed a new APW path for the Guiana Shield covering the time interval between 2155 Ma and 1970 Ma. To construct their APW path, Théveniaut et al. (2006) have favored the amphibole ^{40}Ar–^{39}Ar ages to date paleopoles, in detriment of biotite ^{40}Ar–^{39}Ar ages, since amphibole closure temperatures are normally more similar to magnetite blocking temperatures. Thus, new ages were attributed to paleopoles that defined the APW path traced by Nomade et al. (2003). The GUY1 and GUY2 poles were considered older than 2100 Ma based on geochronological determinations and also using the polarity ambiguity for Precambrian rocks that permit the use of the anti-pole.

Fig. 10 presents the Guiana Shield's APW path traced by Théveniaut et al. (2006) between 2070 Ma and 1960 Ma, where the Surumu Group (SG) pole is also shown (poles described in Table 2). For this interval, 2020 Ma poles OYA and ARMO are well-dated. Their ages were obtained from the dating of different minerals (zircon, amphibole, biotite) representing different closure temperatures. The ages of magnetization were estimated from cooling paths of the different units considering the unblocking temperature of the magnetic carriers. Ages for the other poles are less well-constrained.

A group of poles (TUMU, TAMP03, MATA02, APPR2, APPR5, APPR6 and APPR8 in Table 2) fall over Northern South America, which are interpreted as associated with a Late Rhyacian deformation event, dated at 2070–2060 Ma, and related to closure of Island arc basins (Théveniaut et al., 2006). The average for this particular set of poles, which was designated GF1, has an age of 2050–2070 Ma (see D'Agrella-Filho et al., 2011). Thereafter, the path continues to the east, passing through the ca. 2030 Ma ARMO pole (Théveniaut et al., 2006) and OYA pole (amphibole ^{40}Ar–^{39}Ar age of 2020 ± 4 Ma; Nomade et al., 2001). The younger part of the curve is defined by geological units from Venezuela: Imataca Complex poles (IM1, IM2;

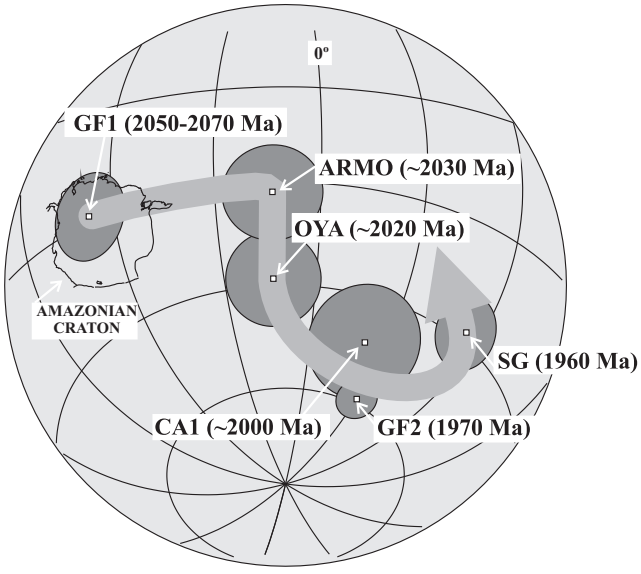


Fig. 10. APW path traced for the Guiana Shield between 2070 and 1960 Ma (after Théveniaut et al., 2006). Amazonia in its present position. Paleomagnetic poles in Table 2.

Table 2
Selected paleomagnetic poles between 2080 Ma and 1900 Ma from Guiana Shield and West Africa Craton.

Continent/formation	Code	Plat (°N)	Plong (°E)	D_p/D_m A95 (°)	Euler Pole[a]	RPlat (°N)	RPlong (°E)	Age (Ma)	References
Amazonian Craton									
a) Surumu Group volcanics	SG	−27.4	54.8	9.8				1960	1
b) Costal Late Granite	PESA	−56.7	25.1	6.2/12.4				~1970	2
c) Costal Late Granite	ROCO	−58.0	26.4	7.9/15.8				~1970	2
d) Costal Late Granite	MATI	−58.6	25.5	9.7/19.4				~1970	2
e) Costal Late Granite	ORGA	−59.7	44.7	10.1/19.5				~1970	2
Mean (b–e)	GF2	−58.5	30.2	5.8				~1970	3
f) Imataca Complex	IM1	−49.0	18.0	18.0				1960–2050	4
g) Imataca Complex	IM2	−29.0	21.0	18.0				1960–2050	4
h) Encrucijada Pluton	EN1	−55.0	8.0	6.0				1969–1972	5
i) Encrucijada Pluton	EN2	−37.0	36.0	18.0				1969–1972	5
Mean (f–i)	CA1	−42.9	21.9	16.5				~2000	1
j) Armontabo River Granite	ARMO	−2.7	346.3	14.2				~2030	2
k) Oyapok granitoids	OYA	−28.0	346.0	13.8				2036 ± 14	6
l) Tumuc Humac Mount. Granite	TUMU	18.9	273.7	19.2/22.3				2050–2070	2
m) Tampok River Granite	TAMP03	−6.9	300.1	15.9/16.1				2050–2070	2
n) Mataroni River Granite	MATA02	14.9	289.2	40.6/42.7				2050–2070	2
o) Approuague River Granite	APPR2	4.5	298.9	19.1/19.2				2050–2070	2
p) Approuague River Granite	APPR5	−5.9	296.9	34.3/35.1				2050–2070	2
q) Approuague River Granodiorite	APPR6	−18.5	294.3	21.3/23.0				2050–2070	2
r) Approuague River Granite	APPR8	5.3	293.4	16.8/17.2				2050–2070	2
Mean (l–r)	GF1	1.8	292.5	11.2				2050–2070	3
West Africa Craton					43.3°N; 330.5°E (−71.5°)				
s) Aftout Gabbro – Algeria	SL	29.0	55.0	6.0		−15.2	5.7	1940–1900	7, 8
t) Harper Amphibolite – Liberia	AH	−10.0	73.0	7.0		−57.9	8.0	1983 ± 34	9
u) Aftout Granite	GAF	−6.0	90.0	8.0		−56.2	40.0	1950–1982	10
v) Ferké Ivory Coast Granite	IC2	−25.0	83.0	16.0		−75.0	20.9	~2000	11
x) Liberia Granulite	OD	−18.0	89.0	13.0		−68.2	39.9	1964–1894	12
y) Abouasi Dolerite	PL2	−50.0	102.0	11.0		−75.1	173.2	2000–2200	13
w) Abouasi Amphibolite	PL1	−56.0	36.0	14.0		−53.3	260.6	2100–2200	13
z) Ivory Coast Granite	IC1	−82.0	292.0	13.0		−32.4	218.3	2085 ± 15	11

Footnote: Refs.: 1 – this study; 2 – Théveniaut et al. (2006); 3 – D'Agrella-Filho et al. (2011); 4 – Onstott and Hargraves (1981); 5 – Onstott et al. (1984); 6 – Nomade et al. (2001); 7 – Sabaté and Lomax (1975); 8 – Piper (1983); 9 – Onstott et al. (1984); 10 – Lomax (1975); 11 – Nomade et al. (2003); 12 – Onstott and Dorbor (1987); 13 – Piper and Lomax (1973). Plat – pole latitude; Plong – pole longitude; A_{95}, d_p, d_m – Fisher's statistic parameters.

[a] Euler rotation pole used to rotate poles and West Africa to Amazonian Craton. RPlat/RPlong – rotated pole latitude/pole longitude.

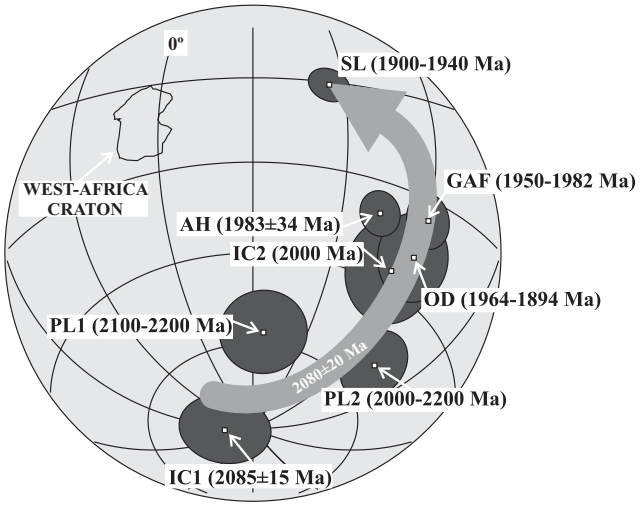


Fig. 11. APW path traced for West Africa Craton between 2085 and 1940 Ma (after Nomade et al., 2003). Africa in its present position. Paleomagnetic poles in Table 2.

Onstott and Hargraves, 1981) and La Encruzijada granites poles (EN1, EN2; Onstott et al., 1984), whose mean is represented by CA1 in Fig. 10 and Table 2, and by a set of four poles (PESA, ROCO, MATI and ORGA in Table 2), whose average is represented by GF2 in Fig. 10. Théveniaut et al. (2006) suggest an age of ca. 1970 Ma for this part of the curve based on the ^{40}Ar–^{39}Ar on amphibole age of 1972 ± 4 Ma obtained from La Encruzijada granites (Onstott et al., 1984). A similar age (ca. 1970 Ma) is suggested for the Imataca Complex, based on its thermal history, inferred from hornblende, biotite and feldspar ^{40}Ar–^{39}Ar datings (Onstott et al., 1989). The Surumu Group (SG) pole is also shown in Fig. 10. An age of 1960 Ma can be attributed for this pole, according to the available U-Pb geochronological ages between 1980 and 1960 Ma for these volcanic rocks (see topic 2), and thus it may represent an extension of the drift curve traced for Guiana Shield for younger ages.

8.3. Paleoproterozoic APW path for the West-Africa Craton

Paleoproterozoic paleomagnetic poles from West Africa are few and poorly dated, $^{40}Ar/^{39}Ar$ and U-Pb datings being practically absent. Fig. 11 shows the APW path for the West Africa Craton between 2080–1900 Ma, using the poles compiled by Nomade et al. (2003) (Table 2). The ages of most paleomagnetic poles are estimated through lithostratigraphic correlations, resulting in large errors (Nomade et al., 2003). The authors suggest an age of 2085 ± 15 Ma for pole IC1 (Fig. 11) determined for granites of a TTG Suite from Ivory Coast (Nomade et al., 2003), based on a regional thermal event that affected the area and was responsible for the NW-SE crustal shortening (D2). The West-African APW path continues northward passing through poles PL1 and PL2 determined for dolerites (PL2) and the Obouasi amphibolites (PL1) (Piper and Lomax, 1973). These rocks are not dated, and large time intervals are associated to them (Table 2). However, their magnetic components are compatible with a VGP determined for a rhyolite in the Ferké volcanic basin, Ivory Coast (site 19, see Fig. 11b of Nomade et al., 2003). Based on stratigraphic constraints and on coherent Rb–Sr and U-Pb ages of other rhyolites in the same region (Lemoine, 1988; Leake, 1992), Nomade et al. (2003) have attributed an age between 2100 and 2060 Ma for this part of the APW path (Fig. 11).

Further to the north, poles OD, IC2, GAF and AH (Table 2) form a cluster with ages between 1960 Ma and 2000 Ma (Nomade et al., 2003). Pole OD was obtained for granulites of Liberia (Onstott and Dorbor, 1987), and the hornblende and biotite from one of the studied rocks yielded ^{40}Ar–^{39}Ar ages of 1964 ± 4 Ma and 1894 ± 2 Ma, respectively. Pole IC2 was obtained for granites collected in Ivory Coast (Nomade et al., 2003). These authors ascribed it an age of ~2000 Ma based on the comparison with other poles of similar ages. Pole GAF was obtained for the Aftout granites (Lomax, 1975), for

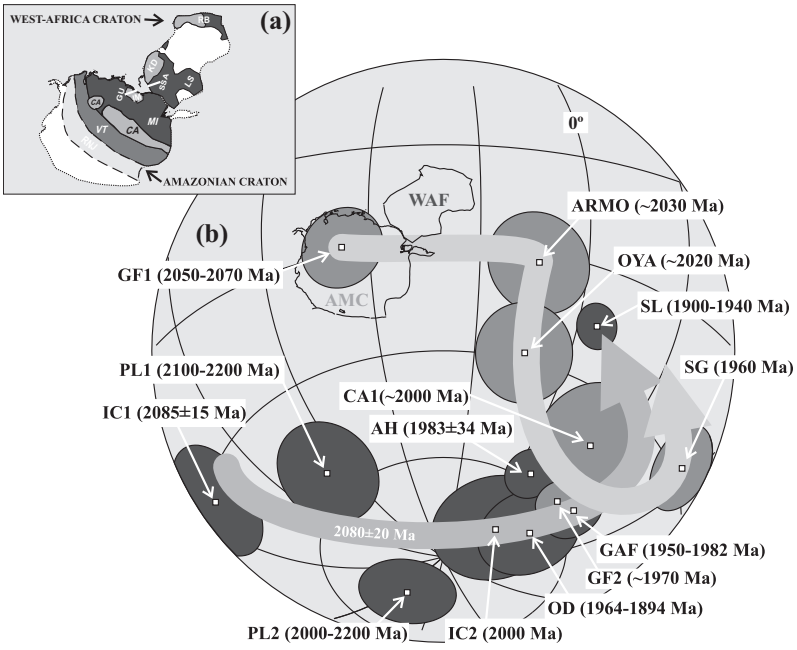


Fig. 12. (a) Paleogeographic configuration of proto-Amazonian Craton (Guiana Shield) and West Africa Craton at about 1970 Ma ago. The Amazonian Craton in its present position, and West Africa Craton was rotated −71.5° about an Euler pole at 43.3°N; 330.5°E. Geotectonic provinces: Amazonia (CA – Central Amazonia Province, MI – Maroni-Itacaiúnas Province; VT – Ventuari-Tapajós Province; RNJ – Rio Negro-Juruena Province, GU – Guri lineament); West Africa (LS – Leo Shield, KD – Kenemanan Domain, RB – Requibat Shield, SSA – Sassandra lineament). (b) Comparison of the (2070–1960 Ma) APW paths from Guiana Shield and West Africa Craton. West African poles in this APW path were rotated using the Euler rotation pole cited above.

which Nomade et al. (2003) assigned an age around 1950–1982 Ma based on stratigraphic evidence. Pole AH corresponds to amphibolites collected near Harper (Liberia) (Onstott et al., 1984). Rb–Sr determinations provided an isochron age of 1983 ± 34 Ma (Hurley et al., 1971) for gneisses correlated to the studied amphibolites, which is similar to a hornblende ^{40}Ar–^{39}Ar age of 1964 ± 2 Ma, obtained for the investigated rock (Onstott et al., 1984). Finally, the northern tip of the curve is defined by the pole SL (Sabaté and Lomax, 1975) determined for the Aftout gabbros for which Piper (1983) assigned an age of 1900–1940 Ma.

8.4. Paleoproterozoic Amazonia/West Africa link

Several studies have suggested that the proto-Amazonian Craton (Guiana Shield) and the West-Africa Craton were linked at the end of the Itacaiúnas-Maroni event (Transamazonic)/Eburnian around 2050–2000 Ma (Onstott and Hargraves, 1981; Ledru et al., 1994; Nomade et al., 2003; Pesonen et al., 2003; Johansson, 2009; Evans and Mitchell, 2011; D'Agrella-Filho et al., 2012; Bispo-Santos et al., 2014, in this issue). Onstott and Hargraves (1981) were the first to suggest a paleogeography where the Guiana Shield and the West Africa Craton were united keeping the continuity of lineaments Guri (Guiana Shield) and Sassandra (West Africa Craton). Paleoproterozoic paleomagnetic poles determined for the two cratonic blocks presented a better fit in this new configuration than in a Gondwana fit. Subsequently, Nomade et al. (2003) showed that the APW paths constructed for the Guiana Shield and West-Africa Craton differed for ages around 2080–2040 Ma, but coincided for ages around 2000–2020 Ma, placing a maximum age for their assembly. In Fig. 12a the Amazonian Craton is shown in its current position and the West Africa Craton is placed after rotation of −71.5° around an Euler pole located at: 43.3°N, 330.5°E. In this configuration, Guri and Sassandra shear zones are aligned as suggested by Onstott and Hargraves (1981) and Johansson (2009), based on geological correlations. Fig. 12b compares the APW paths for the Guiana Shield (Fig. 10) and West Africa Craton (Fig. 11) using the same rotation. Despite the low quality of paleomagnetic poles, the two APW paths are clearly distinct prior to ca. 2000–1980 Ma, but match relatively well for ages around 1980–1960 Ma.

9. Conclusions

This work presents a paleomagnetic study performed on volcanic rocks of the Surumu Group located in the northern part of the Amazonian Craton (Guiana Shield) and dated by U-Pb method between 1980–1960 Ma. The calculated paleomagnetic pole (SG) is located at 234.8°E, 27.4°N (N = 20, α_{95} = 8.9°, K = 12.1°). The SG pole presents reliability factor Q = 5 being considered a robust pole for the Guiana Shield (Amazonian Craton). An inverse positive baked contact test performed for a Mesozoic dyke cutting the Surumu volcanic rocks shows that the Surumu rocks did not suffer recent remagnetizations. The SG pole allowed refining the APW path of the Guiana Shield and support a link with the West-Africa Craton around 1970–2000 Ma ago, in a configuration where the Guri (Guiana Shield) and Sassandra (West Africa) lineaments were aligned.

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